

AI offers way to image and assess clinical cell samples

Yinuo Xu & Zhi Huang

The use of artificial intelligence in the clinic to image slides of cells enables the rapid assessment of cellular characteristics that might help with diagnostic decisions. **See p.472**

Cellular-level information is routinely used as part of clinical diagnostics¹. Laboratory workers use a microscope to inspect cells on glass slides to detect early signs of cancer in organs such as the lung and bladder. Although this method, called cytology, is a long-established diagnostic procedure, it has major weaknesses. Human interpretation is subjective and labour intensive, and workforce shortages are increasingly a challenge². On page 472, Nitta *et al.*³ present an artificial-intelligence approach that offers advantages over current methods in clinical cytology.

There is an urgent need to develop automated workflows that, once shown to be reliable in routine practice, can help laboratories to manage rising demand for diagnostic decisions amid persistent staffing limitations⁴. Such an approach might also enable automated prescreening before a slide is assessed by a clinician and help to improve the consistency of diagnoses.

Automating cellular analysis is not easy: one of the technical difficulties relates to the effectiveness of 3D imaging. Cytologists often refocus slides to examine a sample's various focal levels to clearly inspect features, such as nuclear contours, how cells overlap in clusters and the texture of the nuclear DNA and protein complex known as chromatin.

When such data are interpreted by a person, this brings variability owing to human factors. Even knowledge about a sample can bias a person's interpretation. For example, when examining slides to screen for potential cervical cancer cells, knowing whether a sample is from someone with human papillomavirus infection – the main cause of cervical cancer – can bias a viewer's assessment⁵. Moreover, digitizing images on a slide to capture a variety of focal planes lengthens the image-scanning time and increases the amount of data, creating barriers to easy routine data storage, transfer and AI-model training⁶.

Nitta and colleagues describe an automated system designed to transform cytology workflows, by scanning entire slides in 3D and using AI to diagnose signs of disease without

requiring human prescreening (Fig. 1). The authors made this step forward by optimizing a unified workflow pipeline encompassing imaging hardware, data handling and disease diagnosis. The authors used an imaging approach called whole-slide edge tomography, which enabled them to rapidly acquire images by sweeping the focal plane of imaging for up to 40 focal layers. The computer then reconstructed and compressed images from each frame of the 3D stack using video-style encoding, which reduces the file size but preserves sufficient image quality to provide the details needed for diagnosis. The compressed image stacks are then 'stitched' together to generate a real-time 3D viewing tool that enables the person analysing the sample to easily refocus the frame of the image that they are viewing.

Automation to aid diagnostic decisions

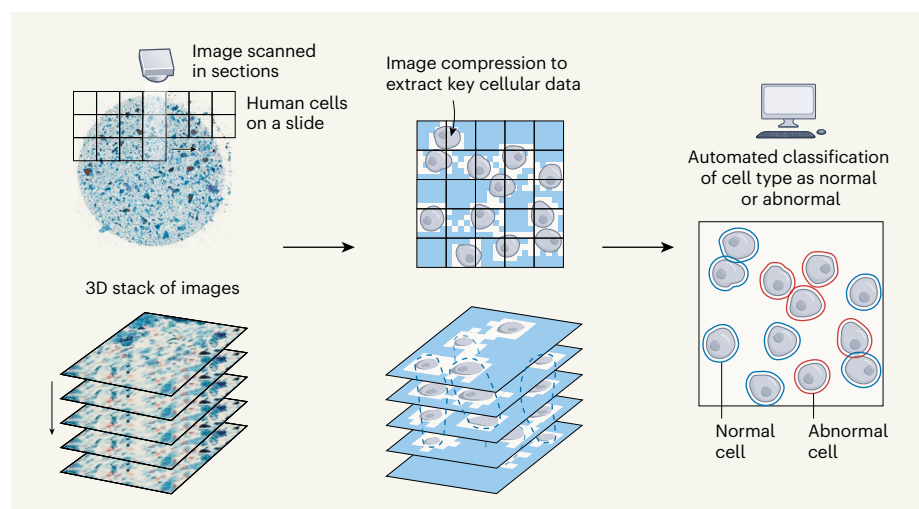


Figure 1 | An AI method for automated assessment of clinical cell samples. Nitta *et al.*³ present a system that can analyse human cells on glass slides to assess whether there are signs of abnormalities, such as cancer or precancerous cells. The system scans the slides over various focal-plane layers, and compresses the image files across and between layers to swiftly 'stitch' together a 3D stack of images that is available in real time for the viewer to scroll through. File compression helps to avoid the collection of large data files that slow processing speeds. The system assesses cells, finding a focal plane in which the nucleus is clearly visible, and then evaluates nuclear characteristics and cellular clustering to predict whether cells are normal or show signs of early or advanced cancer. This type of automated prescreening tool might help people examining such slides. (Adapted from Fig. 1 of ref. 3.)

depends on achieving effective interpretation at the level of individual cells at the whole-slide scale. Nitta and colleagues' system detects nuclei across various focal planes, selects the best-focused plane per nucleus and crops standardized single-cell images for AI classification of cellular status. An AI tool called a neural-network-based classifier assigns each such cell to one of the possible cellular categories. These include normal types of epithelial cell and key abnormal classes of epithelial cell associated with cancer, including low-grade squamous intraepithelial lesions, high-grade intraepithelial lesions and adenocarcinoma. In single-cell tests, the model achieved a near-perfect ability to discriminate between abnormal categories. When aggregating single-cell predictions into slide-level data, the system provides a quantitative census of cell types and the level of abnormal cells present.

How the cellular census is summarized in this work is a particularly interesting development. Nitta *et al.* classified cells using a metric called a cluster of morphological differentiation, in which each cell is represented by a set of numbers describing how likely it is to belong to a cell type based on its appearance. Cells with similar appearance profiles are grouped together, much as certain protein 'markers' are used to classify cells in other types of diagnostic test. When the data are displayed in graphs similar to those used to present results obtained by a technique (called flow cytometry) to separate cell populations, this enables non-informative cellular debris and immune cells called leukocytes to be removed easily from the analysis.

From the archive

Mutant insects could detect contraband, and ash seeds feature a screw propeller

50 years ago

Trained dogs have an established role as aids to Customs Inspectors and Airport Security Guards because of their ability to “smell out” concealed drugs or explosives or other forms of contraband. What seems to have been overlooked is the possibility of using insects for the same purpose. The simpler insects have genetically fixed responses to specific odours, usually resulting in a conspicuous clustering in the vicinity of the odour source which may be of an extremely low absolute intensity. Kikuchi ... isolated a mutant strain of *Drosophila melanogaster* that was strongly attracted to eighteen chemicals which were repellent to the parent strain. It seems likely, therefore, that a systematic search for other strains selectively responsive to various narcotic or explosive emanations can uncover specific “olfactory mutants” of a practically useful sort. While there would be obvious objections to using such species as cockroaches, bees or houseflies, the relatively inconspicuous but not unfamiliar fruit flies would probably be an acceptable alternative to some of the existing security checks.

From *Nature* 11 March 1976

150 years ago

Now that the question of the best form of the screw as a propeller has become of such importance it is interesting to note what Nature has done in this direction. The seed of the ash (*Fraxinus excelsior*) is provided with a wing very delicately twisted, and, when the seed falls, the action of the air upon this screw-like wing causes it to revolve rapidly. The result is that the seed is kept suspended in the air for a comparatively long time, and is wafted by the slightest breeze to a considerable distance from the parent tree ... I should much like to know if any of your readers have observed this screw and studied its pitch, and it would be very remarkable should it prove that the pitch of this natural screw is the one which will give the most power to the propeller of a steamer.

From *Nature* 9 March 1876



The resulting data show how each cell is classified, and visually similar cells appear close together in a 2D map of the results. This map provides a way to assess the similarity between cells on the slide. Such population-level views of cells offer an interpretable step between assessment of images and final diagnostic decisions, and could help with decisions such as whether a slide should be flagged for closer review because it shows signs of disease. This information might also help users to determine whether the AI system makes mistakes.

To assess the potential clinical utility of this tool, the authors assessed whole-slide performance and robustness of the results across 1,124 slides of cells being tested for cervical cancer, obtained from 4 clinical centres. Using the number of abnormal cells per slide as a clinical predictor, the system distinguished, with high accuracy, slides with any cellular abnormality at or above low-grade disease designations (including low-grade squamous intraepithelial lesions, high-grade squamous intraepithelial lesions and a cancer called squamous cell carcinoma) from slides containing normal cells. The system also identified slides of cells with high-grade disease designations (high-grade squamous intraepithelial lesions or squamous cell carcinoma).

Abnormal-cell numbers also provided biological insights. And slides associated with human papillomavirus infection tended to contain more AI-flagged low- and high-grade precancerous cells than did slides not associated with such infections.

An analysis of erroneous predictions by this AI system illustrates an important practical point. In one centre with a relatively younger test population, a benign phenomenon related to cells called navicular cells contributed to results, erroneously suggesting the presence of disease (false-positive results), until the system was revised to take this into account.

There were several advantages to treating image acquisition, data handling and cellular assessment for disease as a single pipeline. On-device image compression addresses the practical bottleneck of enormous volumes of data that has repeatedly limited the use of AI in cytology⁶. Other imaging methods, such as the use of a system called a multi-camera array scanner, can also speed up the 3D digitization of images⁷. In Nitta and colleagues' approach, image acquisition is immediately integrated with storage, streaming and analysis workflows. This makes the system resemble an automated screening tool rather than a system that simply supplements the judgement of experts.

The authors' work highlights key aspects to consider when evaluating autonomous cytology systems. The performance of Nitta and colleagues' system was evaluated against routine cytological diagnoses, which involve the assessment of individual cells collected by

gently brushing a tissue surface or from body fluids in which cells naturally float, without the need to remove tissue, and outcomes were confirmed by histological analysis of a piece of tissue cut from the body. For the new method, as is also the case in routine cytology using current methods, discrepancies in results might arise owing to factors such as sampling variability or ambiguous cellular appearance – as well as any AI-associated error.

Future studies could examine whether automated image assessment affects subsequent clinical workflows and outcomes. Furthermore, the platform's quantitative, cell-level output data might help to monitor clinical information over time, and enable comparisons of data from clinical sites.

Looking even further ahead, this method could be used to enable fundamental biomedical discoveries, as well as to provide diagnostic innovations. As gradual transitions between cell states become visible in images of cells, this information could reveal early processes in disease progression that would be hard to capture with predetermined categorical labels alone. The authors also suggest that the method could be used to examine the link between imaging and the molecular profiling of cells. The method of image-activated cell sorting can physically isolate cells on the basis of their appearance, enabling direct assessments of the relationship between cell shape and genomic or protein characteristics^{8,9}.

Cytology has long relied on humans peering down a microscope. Nitta and colleagues' work suggests that a computer can provide immediate cellular diagnosis at scale. If these findings are confirmed and validated widely in clinical settings, this method might make a cornerstone of cancer prevention even faster, more objective and more accessible.

Yinuo Xu is in the Department of Computer and Information Science, University of Pennsylvania, Philadelphia, Pennsylvania 19104, USA. **Zhi Huang** is in the Department of Pathology and Laboratory Medicine, University of Pennsylvania, Philadelphia, Pennsylvania 19104, USA.
e-mails: xuyinuo@seas.upenn.edu;
zhi.huang@pennmedicine.upenn.edu

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